1. Introduction

1.1 Overview

Lung cancer is a major contributor to cancer-related deaths globally, making up a large fraction of cancer fatalities each year [1, 2]. Timely and accurate diagnostic methods can be lifesaving, as early detection leads to appropriate and targeted treatment [3]. Many types of medical imaging technologies like CT scans and chest x-ray are employed to facilitate correctly disease diagnosis. Recent advancement in artificial intelligence [4], particularly in deep learning [5], enables the automation of medical imaging analysis with exceptional accuracy [6]. Convolutional neural networks (CNNs) [7] are particularly efficient at identifying complex visual features, which makes them useful for classifying medical images. Deep learning models often produce good results. However, people treat them as black boxes [8]. This is because these models give outputs without any explanation. So, understanding their decision-making process becomes very difficult. For this reason, researchers are exploring Explainable Artificial Intelligence (XAI) in medical imaging research because it stands out in terms of model interpretability [9]. The main goal of it is to make AI models easier for humans to understand [10]. In health care, this factor is extremely important since doctors need to understand the rationale for making decisions in order to be able to trust the system.This factor is extremely important in health care, since doctors have to understand the rationale behind the decision made. The most used approach for interpretability is Grad-CAM (Gradient-weighted Class Activation Mapping) [11]. It creates a heatmap that highlights the key areas of an image. These areas are the ones the model focused on to make its prediction.

This study develops a deep learning-based algorithm to classify lung images into three categories: normal, benign, or malignant. At the same time, explainable AI algorithms will be applied to produce interpretable visual explanations of predictions.

1.2 Background

Lung cancer poses a significant public health problem as an uncontrollable cellular growth in the lungs' tissues [12]. There exist two categories of conditions – benign and malignant. The latter is associated with cancer development, whereas benign lung disease involves non-cancerous abnormalities that may also require clinical treatment [13]. Survival chances are increased when a condition is diagnosed early, thus accurate classification of a lung disease is critical for effective diagnosis and Medication [2]. Diagnosis of lung cancer depends on medical imaging since it allows physicians to detect signs of disease in the lungs. Yet, the interpretation of the images may be challenging and subjective. Hence, computer-aided diagnosis (CAD) systems were introduced for automatic image analysis that would help radiologists make decisions based on visual examinations [6].

Deep learning has seen major progress recently, especially with CNNs. This has greatly improved automated image classification [14]. The algorithm is able to correctly detect input images into benig, malignant and normal cases. However, they lack the explainability [15]. Explainable Artificial Intelligence provides transparency concerning the prediction results obtained via neural networks. Grad-CAM produces the image that demonstrates the region of the input image that is accountable for making the prediction [11]. It is important for the medical application domain because AI should provide reliable solutions to health specialists [16].

1.3 Problem Domain and Motivation

Lung cancer diagnosis based on medical imaging poses several difficulties owing to variations in the dimensions, shape, texture, and location of lesions [17]. Radiologists apply their knowledge and skills in interpreting the image results, which can be a lengthy process and prone to bias [18]. Automated models using deep learning technology can help by providing reliable and fast classification [6]. Despite their excellent results, deep learning models generally lack interpretability, thus limiting their implementation in a real-world environment [19]. An interpretability-free model can prove unreliable in crucial fields like medicine because it does not provide an explanation for its decision-making processes. The lack of interpretability leads to a discrepancy between the model's actual performance and applicability [20]. In addition, the differentiation between malignant and benign cases and normal conditions is necessary for accurate diagnosis and treatment planning since misclassification will cause late treatments or unnecessary operations. Therefore, there exists a necessity for a model that accurately classifies lung cancer cases and provides interpretable results.

The motivation behind this research is to develop a diagnostic tool that incorporates explainable artificial intelligence methods. This technology allows users to

generate explanations for model decisions graphically.

1.4 Research Questions

We developed the following research questions to guide this study.

- Is it possible for a deep learning system to correctly detect lung images as benign, malignant, or normal?
- How can XAI help us interpret the outputs of a model?
- Does the interpretability provide any valuable understanding about the model's reasoning?
- What are the performances of the proposed model?

1.5 Objectives of the Thesis

This thesis aims first to develop a deep learning system. That can interpret medical scan results for lung cancer detection. These specific objectives include:

- Designing a CNN based system that will able to classify lung images as either normal, benign, or malignant.
- Adopting explainable artificial intelligence (XAI) to visualize and explain how the model arrives at its predictions.
- Check how well the system works. This will be done with standard metrics including precision, accuracy, recall, and F1 score

1.6 Thesis Outline

The rest of this thesis is organized as follows. Chapter 2 gives an overview of past studies on medical imaging and machine learning for lung cancer diagnosis. It pays special attention to explainable artificial intelligence and how it matters for medical diagnosis. Chapter 3 describes how this study was carried out. It covers the dataset, the steps taken to prepare the images, how the model was trained, and the explainability method used. Chapter 4 explains the experiments that were run. It also presents the metrics used to judge the results, the analysis of the data and model performance, and how Grad-CAM was applied for visualization. Finally, Chapter 5 sums up the main findings of this research. It lists the key contributions, points out the study's limitations, and offers possible directions for future work.

2. Literature Review

2.1 Literature Review

Deep learning has an outstanding ability for detecting lung cancer using medical imaging data [21]. Many existing research studies try to increase accuracy and reduce false alarms for automating the process of diagnosing.

Initially, most researches have utilized convolutional neural networks for classifying lung nodules. For example, Joshua et al. [22] suggested using a simple 3D CNN model along with Grad-CAM to classify lung nodules with an accuracy rate of 97.17%. In another study, Saravanaprasad et al. [23] utilized a 3D CNN along with an SVM model as part of their computer-aided diagnosis system, with an accuracy rate of 94%. This shows how deep learning can help detect informative features from medical images. Nevertheless, most of these models are computationally intensive and have been trained using a limited number of data samples.

To further improve performance, several studies introduced hybrid and multi-stage models. Hamidreza et al. [24] combined GAN, reinforcement learning, and deep networks for better segmentation and detection. Wankhade et al. [25] developed a hybrid system. It uses both 3D-CNN and RNN to tackle many tasks inside a single framework. In a similar way, Naseer et al. [26] have used customized CNN models with SVM classifiers, with results of 97.98% accuracy. Though such models offer highly accurate results, they increase the cost associated with training complex models.

Segmentation and classification together have also received some attention from researchers. Crasta et al. [27] used a combined 3D-VNet and 3D-ResNet architecture, offering 99.2% accuracy in terms of classifying nodule type. Another interesting approach by Yahia et al. [28] is using a transformer-based UNETR model for segmentation, along with classification. Such methods can improve detection, but at the same time, they need more computational power.

Zhang et al. [29] have designed a multiscene deep learning architecture for better detection and reduced number of false-positive cases. On the other hand, Asuntha et al. [30] designed a hybrid model for lung nodule detection and its severity analysis through the combination of CNN with optimization. However, there are several drawbacks of these approaches because they consist of several

steps and are not end-to-end. Besides, researchers have used ensemble methods, as well as multiple models with multiple modalities in recent studies. Emad et al. [31] used a dual-branch approach involving image and mask, showing 98.04% accuracy. Also, Saha et al. [32] developed a stacked transfer learning method using CNN models for classification purposes.

Additionally, the issue of efficiency was considered in a number of research papers. Specifically, Haziq et al. [33] designed a sequential CNN model that is efficient and has an accuracy of 95.34

Moreover, Abe et al. [34] developed an ensemble CNN model that is able to achieve 98.17% of accuracy in detecting early-stage lung nodules on the basis of CT scan images. As a result, it is evident that ensemble deep learning systems are more efficient in helping radiologists to detect tumors.

Rai et al. [35] used deep learning for lung cancer screening using the IQ-OTH/NCCD [36] dataset and compared a CNN to several transfer learning methods. By applying SHAP interpretation techniques, the authors obtained 97.3% accuracy of results, stressing the importance of balancing precision and recall scores for reliable predictions.

Elhassan et at. [37] suggested a two-stage approach which involves classification based on CNN and tumor detection using YOLOv8 and augmented with the use of DCGAN technique. The method reached 97.67% of accuracy and an IoU score of 0.85. However, this technique requires a higher degree of methodological complexity and lacks the ability to classify several classes and produce interpretable models.

Moreover, there are a number of review articles about research on lung cancer detection and diagnosis. For instance, Gao [38], Wang [39], and Tan [40] note that CNN-based approaches dominate in this area, while also mentioning several problems like the lack of standardization, small samples, and poor generalization abilities. Moreover, according to Liu et al. [41], meta-analysis shows that AI-based models demonstrate high accuracy of diagnostic results.

Nevertheless, there are some difficulties that need to be addressed. Many of the available literature focuses on enhancing accuracy without providing any interpretation or explanation of their results. Lack of interpretability is a major barrier towards the adoption of such methods in medical environments since they need to be highly transparent. Additionally, many papers focus on binary classification which is not enough for real-life applications. In addition, many models rely on complex architectures or multi-stage pipelines, making them difficult to deploy. Limited dataset diversity and lack of external validation also affect the generalization of these models.

Therefore, we need a deep learning system. It must perform well at multi-class lung cancer detection and provide interpretable outputs to support clinical decision-making.

Table 2.1: Summary of selected recent studies on lung cancer detection

Author (Year)	**Method**	**Dataset**	**Task**	**Result & Strength**	**Limitations**
Crasta (2024)	3D-VNet + 3D-ResNet	LUNA16	Seg+Cls	99.2% acc; robust performance	No XAI, binary focus
Saravanaprasad (2024)	3D CNN + SVM	Kaggle CT	Cls	94% acc; effective CAD	No interpretability, not multi-class
Hamidreza (2024)	GAN + RL + DL	LIDC-IDRI	Seg+Det	IoU 0.97; strong detection	Complex pipeline, no XAI
Eldho (2024)	3D-DLCNN + Mask-RCNN	Clinical CT	Det+Sev	93% acc; low false positives	No interpretability
Saha (2024)	Ensemble CNN	Kaggle CT	Multi-cl	91% acc; stable classification	No XAI, high complexity
Tyagi (2023)	3D Dense CNN	CT datasets	Staging	97% acc; detailed staging	Complex model, no XAI
Wankhade (2023)	3D-CNN + RNN	LIDC-IDRI	Multi-tk	Multi-task capability	No explainability, unclear validation
Yahia (2023)	UNETR (Transformer)	MSD CT	Seg+Cls	98.7% acc; advanced architecture	High complexity, no XAI
Naseer (2023)	U-Net + AlexNet + SVM	LUNA16	Pipeline	97.9% acc; full pipeline	Multi-stage, not interpretable
Haziq (2025)	Sequential CNN	Histopath.	Cls	95% acc; efficient model	No XAI, limited class scope
Emad (2026)	CNN + DNN (Dual-branch)	LIDC-IDRI	Multi-mo	98% acc; strong fusion	Complex, no interpretability

3. Methodology

3.1 Introduction

This study aims to build an accurate deep learning model for lung cancer detection. The model will also use explainable AI to provide interpretable results. Deep learning methods are now widely used for analyzing medical images. One reason is that they can pull out complex features. Another reason is their high accuracy [42]. Diagnosing lung cancer in CT images presents various challenges, such as the variability in cancer size, shape, density, and location [43, 44]. Furthermore, manually diagnosing cancers can be quite challenging and time-consuming. Deep learning has proven highly effective for diagnosing cancer from medical images [45]. This is especially true for convolutional neural networks (CNNs), which perform very well at this task. Unfortunately, when deep learning methods do not offer clear explanations, they may become too unreliable for use in clinical settings. To address this limitation, XAI techniques are increasingly being integrated with deep learning models in the medical decision-making process [46, 47].

Our proposed system will consist of multiple stages of data processing, including preparation, pre-processing, model training, prediction, and explanation. Initially, images from CT scans will be collected and categorized into specific groups. Following this, pre-processing steps will be undertaken to make the images clearer and better for model training. Then, we will use the dataset to train a deep learning system that can detect lung cancer. Finally, explainability will be attained through the use of XAI techniques.

The primary objectives of this methodology consist of the following:

- To propose a system for detecting lung cancer through CT scan images.
- To detect lung images in three categories: benign, malignant, and normal.
- To enhance the precision of models by utilizing effective preprocessing techniques.
- To apply explainable artificial intelligence for interpreting the model's outputs.

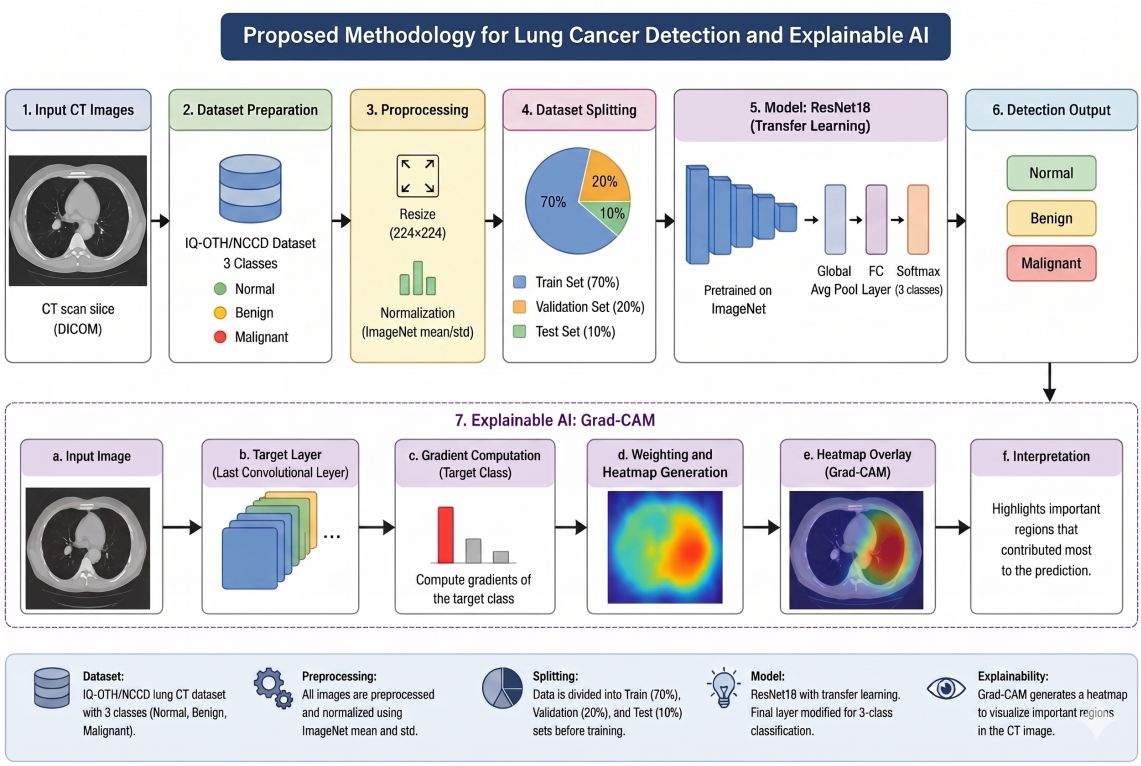


Figure 3.1: Proposed methodology of lung cancer detection system with explainable AI.

3.2 Dataset

For our study, we used a publicly available dataset called IQ-OTH/NCCD [36] to train, test, and validate our lung cancer detection system. This dataset is used in existing research as a benchmark. The dataset used in this study comes from two sources in Iraq: the Iraq Oncology Teaching Hospital and the National Center for Cancer Diseases [48] The source of the dataset was the Iraq Oncology Teaching Hospital and the National Center for Cancer Diseases [48]. This dataset contains a total of 1,190 CT scan samples that fall into three groups:

- 55 cases for Normal
- 15 cases for Benign
- 40 cases for Malignant

Each case includes many CT slices, anywhere from 80 to 200 images, showing different cross-sections of the lung. The raw scans were stored as DICOM files. Later, they were changed into standard image formats for easier handling [49].



(a) (b) (c)

Figure 3.2: Some samples of CT images from the dataset where (a) normal, (b) benign, and (c) malignant cases.

However, the dataset is small and has imbalanced classes. This can make it hard for the model to learn properly.

3.3 Data Pre-processing

Data preprocessing steps are very important for clinical data because they increase data quality and make it suitable for deep learning models [50]. The effectiveness of clinical data analysis using deep learning algorithms greatly relies on the effectiveness of the preprocessing process, especially in the medical field [51].

3.3.1 Image Resizing

Images are standardized in size for all the input pictures in order to maintain uniformity in the entire database. This process is necessary since the model used requires consistent size inputs, besides making it easier for computation and hence efficient during training [52, 51]

Let the original image size be $W_{orig} \times H_{orig}$. The image is resized to a target dimension $W_{target} \times H_{target}$.

3.3.2 Normalization

Normalization is used to adjust pixel values so they fall between 0 and 1. It divides each pixel value by 255. It is defined by Eq.3.1.

$$I_{scaled} = \frac{I_{pixel}}{255} \tag{3.1}$$

This process helps in stabilizing the training process and enabling faster, more reliable model convergence [53].

3.3.3 Data Augmentation

To tackle the problems faced with having a small training set and class imbalance, methods such as data augmentation were implemented. Data augmentation increases the size of the training set artificially and may help reduce overfitting [41]. The following data augmentation methods were used:

The following augmentation techniques are used:

- Rotation: The image undergoes rotation at a randomly selected angle in a certain range, which ensures that the model is invariant to the orientation of the medical feature.
- The image is flipped horizontally, essentially increasing the training set by a factor of two and preventing the model from being oriented towards a particular direction for diagnosis.
- Zooming: Random zoom in and out functions are performed on the images, which enables the network to learn features at varying distances from the camera to subject.
- Shift: Random horizontal and vertical shifting is performed, which helps the model learn position-agnostic detection of the target region.

3.3.4 Dataset Splitting

After preprocessing, we divided the data into three portions. 70% went to training, 10% to testing, and 20% to validation. After splitting, we applied an augmentation technique to the training set only to ensure that augmented images derived from a single original image are not distributed across both the training and validation sets. Such leakage could lead to overly optimistic performance estimates, as the model may indirectly learn from similar samples present in both sets. We do not touch the testing set. This guarantees a fair evaluation of how well the model handles fresh data it has never seen before.

3.4 Model Selection

Choosing the right model is essential for obtaining dependable results in medical image classification. We trained and evaluated several models alongside our proposed model, including Xception, MobileNetV2, CoAtNet, and an Ensemble

method (Max Voting). In our thesis work, we utilized a transfer learning approach founded on the ResNet18 architecture.

ResNet (Residual Network) is a sophisticated deep convolutional neural network [54]. It incorporates residual connections to address the vanishing gradient problem [55]. This allows deeper networks to be trained more effectively. Among the different versions available, ResNet18 was chosen for this work. It offers an optimal balance between predictive performance and computational efficiency [56].

Transfer learning is not the same as training the model. This technique allows using features that have been learned through pretraining on the big datasets, such as ImageNet [57][58]. The benefit of this approach when applied to medical images is that these images belong to relatively small datasets. With initial pretraining, it becomes easier for the model to learn simple characteristics like edges and textures, which become very useful in the case of CT scans.

Using alternative approaches, like ViTs [59], would require a lot bigger data samples. Therefore, ResNet18 is chosen in this research due to its efficiency and effectiveness when working with relatively small datasets.

In our experiment, we changed only the last layer of the ResNet18. The reason is that this layer makes predictions about three different classes - normal, benign, and malignant. All other layers were left unchanged. In this way, it becomes possible for the model to learn how to classify the images accordingly.

3.5 Proposed Model Architecture

The proposed framework adopts the ResNet18 architecture described in [54]. This architecture includes several convolutional layers equipped with residual connections. A fully connected layer is then appended to perform classification [54].

3.5.1 Residual Learning

The central concept of ResNet revolves around residual learning. The network does not directly predict the output. Instead, it learns the residual. The residual is simply how much the input and output differ from each other. This idea is shown in Eq.3.2.

$$y = F(x) + x \tag{3.2}$$

In this equation, x stands for the input, $F(x)$ represents the residual function, and y is the final output. This helps in preserving gradient flow and improving training stability [54].

3.5.2 Network Modification

The pre-trained ResNet18 model is modified by replacing its final fully connected layer, which is defined by Eq.3.3.

$$y = Wx + b \tag{3.3}$$

Here, W and b are parameters that the model learns during training. The output y has three values, one for each class: normal, benign, and malignant.

3.5.3 Activation and Output

The output of the last layer undergoes a transformation using the softmax function. It is defined by Eq.3.4.

$$P(y = i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \tag{3.4}$$

This enables multi-class classification [60].

3.6 Model Training and Evaluation

We train the model using a preprocessed dataset. It follows basically three steps, such as forward propagation, calculating the loss, and backward propagation.

3.6.1 Loss Function

During training, the model figures out its error using categorical cross-entropy. This is shown in Eq. 3.5.

$$L = -\sum_i y_i \log(\hat{y}_i) \tag{3.5}$$

3.6.2 Optimizer

During training, the AdamW optimizer is employed. This method integrates an adaptive learning rate with weight decay regularization to enhance model performance [61]. It is defined by Eq.3.6.

$$\theta = \theta - \eta \cdot \hat{m} / (\sqrt{\hat{v}} + \epsilon) \tag{3.6}$$

where θ represents model parameters.

3.6.3 Learning Rate Scheduling

We also use a ReduceLROnPlateau scheduler [62]. A reduction in the learning rate occurs once the validation loss stops decreasing. As a result, the model's convergence becomes smoother.

3.6.4 Training Setup

The model undergoes training via mini-batch gradient descent with the parameters listed below:

- Batch size: 32
- Learning rate: 1×10^{-4}
- Epochs: 30
- Input size: 224×224

The model is trained using a GPU when one is available to speed up the computation process. Following the training, we assess the performance of the proposed model with a distinct test dataset that was not utilized during the training or validation phases. Our evaluation employs various metrics, including Accuracy, Precision, Recall, and AUC-ROC [63, 64]. We also use a confusion matrix to see how well the model performs on each of the three classes. Taken together, these metrics show us exactly how good the model is at finding lung cancer among the three types.

3.7 Explainable AI using Grad-CAM

While deep learning models may demonstrate high accuracy when detecting abnormalities based on medical images, they often remain black-boxes. The inability to determine why the model makes particular predictions is a problem. For clinical settings, interpretability is especially important because it helps build confidence in a model and assists medical experts in making well-informed decisions. To ensure our model is interpretable, we use Grad-CAM [11].

Grad-CAM is one of the most used methods in explainable AI for imaging data. It determines the area of the image that significantly influence the prediction of the machine learning model. The method allows us to visualize such regions by creating heatmaps of input images.

To use Grad-CAM, we feed the trained ResNet18 model with the final convolutional layer and generate a heatmap for each CT image input. We then combine this heatmap with the original image, which helps to see what parts of the lungs are particularly attention-grabbing for the model. For instance, in malignant cases, the model usually pays special attention to the suspected nodules or other problematic parts of the image.

There are a few major strengths to using Grad-CAM:

- Helps to visualize the regions influencing the model decision [65].

- Assists in ensuring that the machine learning model focuses on clinically relevant parts of the image.
- Helps medical experts in interpreting the model prediction [66].
- Makes the system more transparent and reliable [67].

With the help of Grad-CAM, the new system is able to detect different types of lung cancer accurately and at the same time provide explanations, hence overcoming one of the main weaknesses of deep learning algorithms in healthcare.

3.8 Financial Planning for the Project

Table 3.1 outlines the estimated budget for the Reliable Lung Cancer detection from Medical Images and Model Interpretability project. This includes key components like personnel, hardware, software, and data acquisition to ensure successful project completion.

3.9 Gantt Chart of our Thesis work

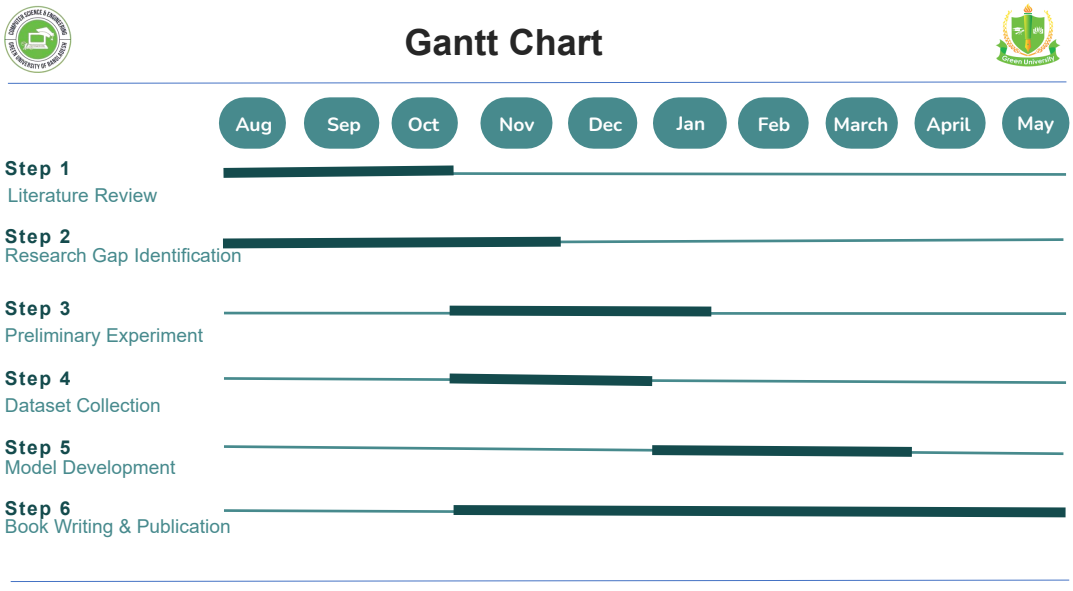


Figure 3.3: Gantt chart of our system development.

Figure 3.3 shows the project schedule over 10 months. It includes six different steps and how long each step is expected to take. It begins with the Literature

Table 3.1: Budget Breakdown for the lung cancer detection with explainable AI.

Budget Type	Description	Unit	Total (BDT)
Hardware	Purchase of storage devices (SSD, external hard drives) for seamless data storing.	02	20,000
Software	Premium membership of Lucidchart and Overleaf for designing methodology, architecture, and writing book.	02	10,000
Data Acquisition	Expenses for acquiring datasets (lung images), annotating them, and applying augmentation for training.	01	30,000
Cloud Computing	Usage of cloud services (such as Colab Premium, Kaggle Premium) for model training and storage of large datasets.	02	20,000
Miscellaneous	Miscellaneous costs including transport, printing, reproduction costs, mobile bills, and small research materials.	01	10,000
Grand Total			**90,000**

Review and Research Gap Identification from August to May. Preliminary Experiments are conducted in August-January to assess feasibility. Dataset collection takes place in November and December to gather the data needed for the next stage. Model development happens from January to March. Finally, thesis writing and publication run from November through May, making sure all findings are properly documented.

3.10 Conclusion

This chapter discusses the detection process used for lung cancer in this study. The use of the IQ-OTH/NCCD [36] dataset with CT scan images classified as normal, benign, and malignant is done. Preprocessing was performed to ready the data for the learning process through a series of steps such as normalization and resizing.

The transfer learning algorithm using ResNet18 as a basis was used for clas-

sifying the CT scan images into three different categories. With limited medical information available, pre-trained weights were used. Optimized learning algorithms and suitable evaluation metrics were used to train and get good results from the model.

Noting that deep learning models can be very difficult to interpret, Grad-CAM was added as a way of making the model more understandable. It helps by highlighting parts of the CT scan image that are contributing to the model decision-making process.

Overall, the method proposed is very efficient in detecting lung cancer.

4. Result Analysis and Discussion

4.1 Introduction

This section outlines the outcomes and analysis of our lung cancer detection system. Our model's performance was assessed with standard classification metrics using the IQ-OTH/NCCD [36] dataset. The results are examined from various perspectives, including the model's training effectiveness, the accuracy of its classifications, and the ease of interpretation.

In this chapter, we evaluate our ResNet18-based model's ability to label CT images as benign, malignant, or normal [68]. Additionally, Grad-CAM is used to assess model explainability. It helps us figure out which areas within an image played a role in the model's final call.

4.2 Experimental Setup

We developed our system leveraging PyTorch, a deep learning framework [62]. The experiments utilized the IQ-OTH/NCCD [69] lung cancer dataset, which comprises CT scans categorized into three classes: benign, malignant, and normal [68]. Prior to the training phase, the images were adjusted to a size of 224×224, and normalized with the ImageNet dataset's mean and standard deviation values. We divided the data into three parts. Training got 70%, validation took 20%, and the last 10% went to testing. This division was carried out before applying any data augmentation to prevent data leakage.

In this research, we employed a ResNet18 model with transfer learning. We initialized with weights that had been pre-trained on ImageNet. Then, we modified the final fully connected layer so it can output three classes.

Training was conducted with the following configuration:

- Optimizer: AdamW
- Learning rate: 1×10^{-4}
- Batch size: 16

- Number of epochs: 30
- Loss function: Cross-Entropy Loss

The ReduceLROnPlateau schedule is used for adapting the learning rate according to the validation loss. If available, the experiment will be done using GPU acceleration.

4.3 Performance Evaluation Metrics

Below is a set of metrics we used to evaluate our model. They show us how efficient the model is, especially for detecting lung cancer in three classes.

4.3.1 Accuracy

Accuracy tells us how often the model gets the answer right overall [70], calculated using Eq.4.1

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{4.1}$$

4.3.2 Precision

Precision shows, out of all the times the model predicted a certain class, how many of those predictions were actually correct [71], as quantified by Eq.4.2.

$$Precision = \frac{TP}{TP + FP} \tag{4.2}$$

4.3.3 Recall

Recall, which is also called sensitivity, measures how many of the real positive cases the model managed to catch [72]. Its calculation formula is provided in Eq.4.3.

$$Recall = \frac{TP}{TP + FN} \tag{4.3}$$

4.3.4 F1 Score

The F1-score brings precision and recall together by taking their harmonic mean as expressed in 4.4.

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \tag{4.4}$$

4.3.5 Confusion Matrix

To evaluate the model for each class, confusion matrices evaluate the accuracy of the model prediction by counting the number of true and false predictions.

4.3.6 Receiver Operating Characteristic (ROC) Curve

This approach evaluates how well the model separates classes' power by plotting the ROC curve for each class using the one-versus-rest method. The AUC [73] then gives us a single number that reflects how good the model is at discriminating between classes.

4.3.7 Explainability using Grad-CAM

Explainable AI highlights specific regions in a CT image. These regions influence the model's predictions the most. This helps verify whether the model focuses on the relevant parts of the lungs.

4.4 Training Performance

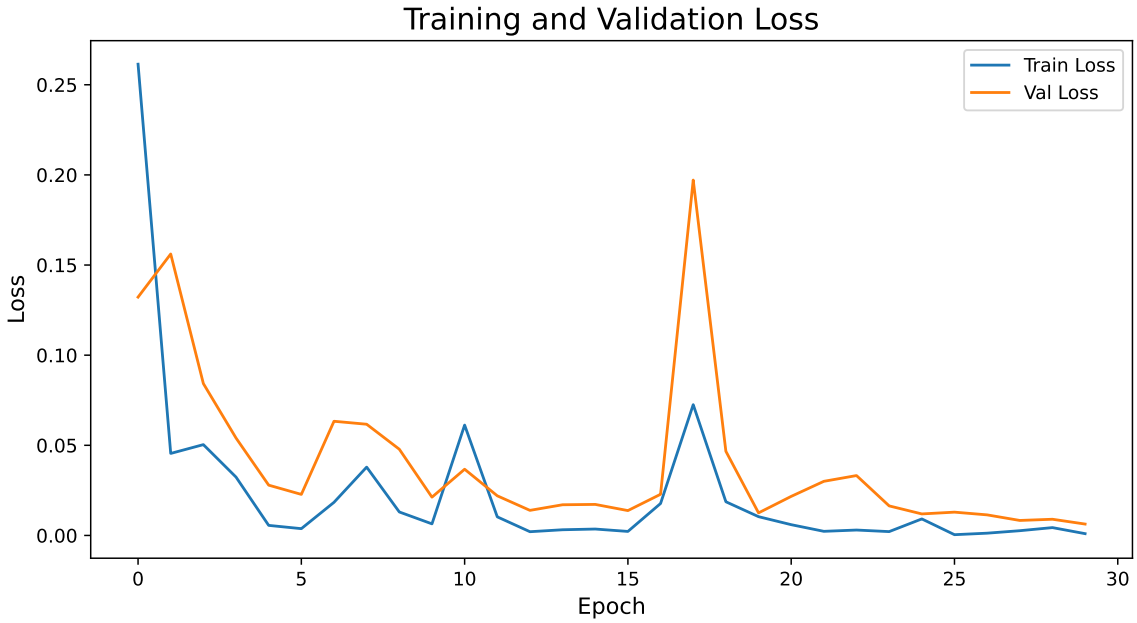


Figure 4.1: Loss curves over epochs show stable convergence of the proposed model

The curves representing the training and validation loss are shown in Figure 4.1. As shown in the graph, both the lines show a consistent decline with respect to

epochs, indicating that the model learns continuously without undergoing overfitting. Therefore, it can be inferred that the selected learning rate and parameters of the optimizer are suitable for the task at hand. Likewise, Figure 4.2 represents the training and validation accuracy curves. The validation accuracy stays close to the training accuracy, which suggests that the model generalizes well to new data [74].

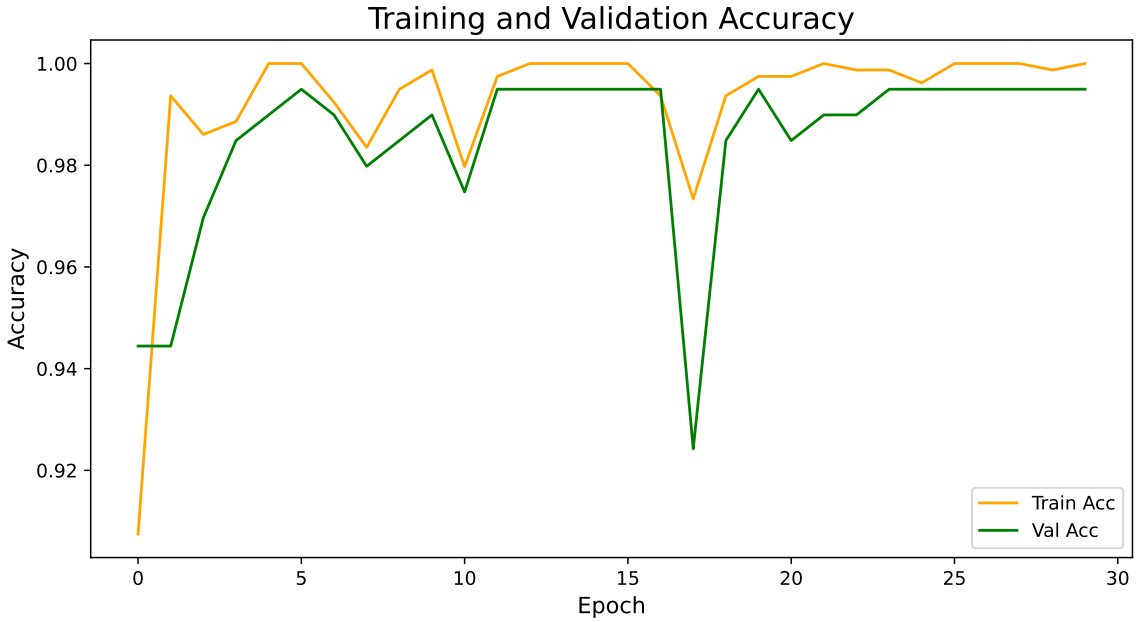


Figure 4.2: Accuracy curves over epoch demonstrating consistent performance and good generalization

4.5 Classification Results

Confusion matrix has been shown in Figure 4.3. The model perfectly classified all malignant and normal images while just one benign image was wrongly classified. This shows that the model performed remarkably well, especially when considering vital categories such as malignant cancer. Correct identification of malignant cancer is very important as far as early diagnosis is concerned [2].

4.5.1 Classification Report

Figure 4.4 shows the classification report. It gives a detailed look at how the model performs. Overall, the model achieves 99% accuracy. However, in terms of recall of benign, a lower score shows that some of the benign cancers were wrongly classified because of class imbalances in the data set.

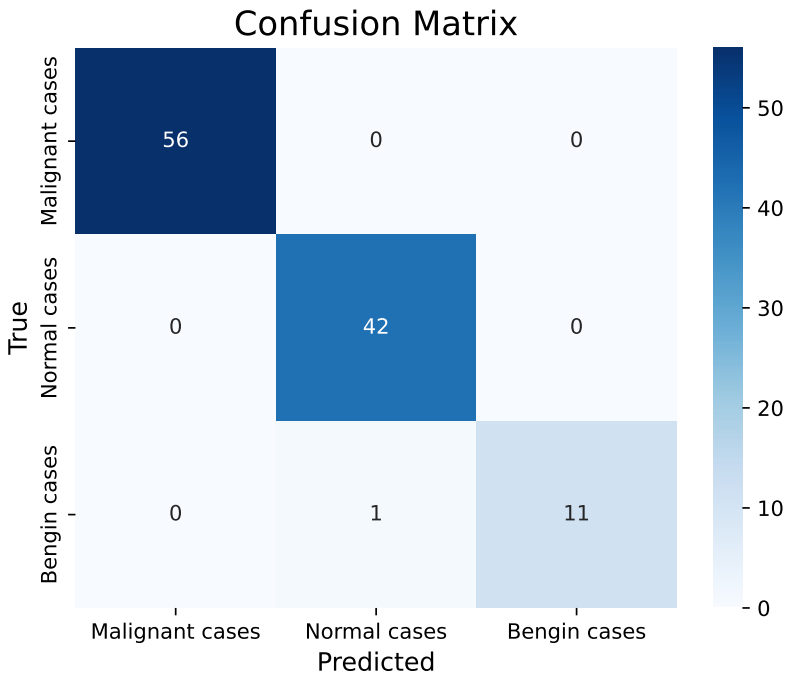


Figure 4.3: Confusion matrix showing how well the model classifies normal, benign, and malignant cases

4.6 ROC Curve Analysis

All three ROC curves are shown in Figure 4.5.

The model achieves excellent AUC scores:

- Malignant: AUC = 1.000
- Normal: AUC = 0.997
- Benign: AUC = 0.993

These results indicate that the model has strong discriminative capability across all classes.

4.7 Explainability Analysis

The visualization outcome using Grad-CAM [11] is presented in Figure 4.6 below. The heat maps emphasize parts of the CT images that have the most significant impact on the decision-making of the model. For a malignancy situation, the model focuses on the unusual lungs that are relevant to the diagnosis process. It

Classification Report:

	precision	recall	f1-score	support
Malignant cases	1.00	1.00	1.00	56
Normal cases	0.98	1.00	0.99	42
Bengin cases	1.00	0.92	0.96	12
accuracy			0.99	110
macro avg	0.99	0.97	0.98	110
weighted avg	0.99	0.99	0.99	110

Figure 4.4: Class-wise proposed model's performance metrics

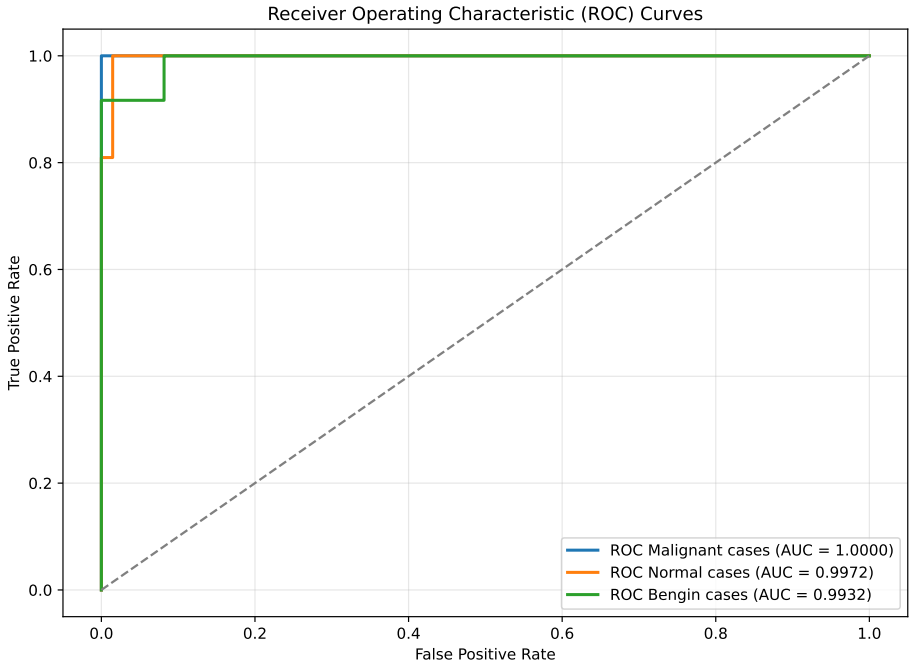


Figure 4.5: ROC curves for multi-class classification showing high AUC values for all classes

shows that the model is accurate and explainable since it makes decisions based on anatomically relevant parts of the CT images [35].

4.8 Performance Comparison with Other Models

For assessing the efficacy of the proposed method, various deep learning models were tried on the same data set. The classification accuracy results of the various architectures are shown in Table 4.1.

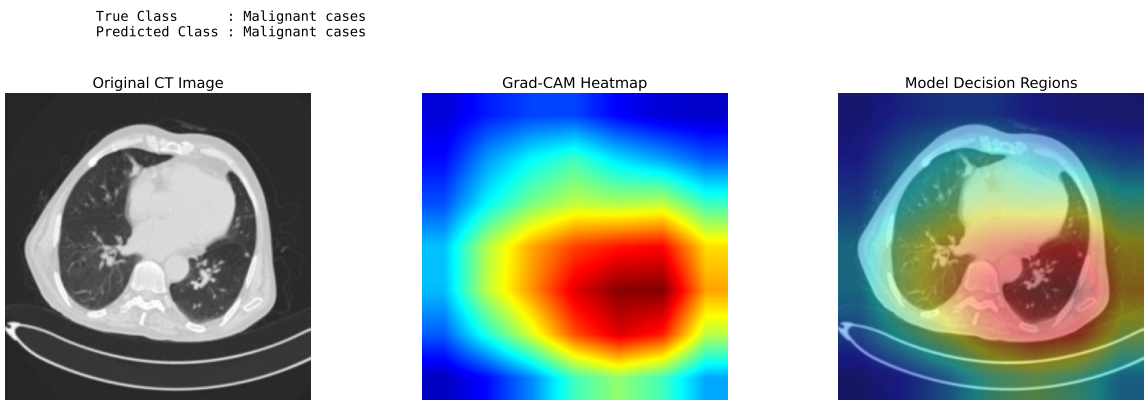


Figure 4.6: Grad-CAM visualization highlighting important regions influencing the model's prediction in CT images

Table 4.1: Comparing how well various deep learning models perform

Model	**Accuracy**
Xception	85.00%
MobileNetV2	82.50%
CoAtNet	98.89%
Ensemble (Max Voting)	93.89%
ResNet50	95.56%
Proposed ResNet18 + Grad-CAM	**99.00%**

From Table 4.1, we see that our proposed model achieves the highest classification accuracy among all models. MobileNetV2 and Xception show comparatively lower performance, while the ensemble and ResNet+ models improve the results significantly. CoAtNet achieves competitive performance with 98.89% accuracy; however, the proposed model slightly outperforms it while also providing explainability through Grad-CAM visualization.

4.9 Comparative analysis with existing works

Table 4.2 shows the comparison between existing work and our proposed work. This comparison can be categorized into three distinct groups.

Firstly, hybrid and complex models such as those developed by Saravanaprasad et al.[23] and Emad et al.[31] achieve accuracy levels ranging from moderate to high (94%–98.04%), but they depend on multi-stage or dual-branch architectures, which lack explainability. Similarly, Abe [34] attains a high accuracy of 98.17% by employing an ensemble CNN, though this results in increased computational complexity and diminished interpretability.

Table 4.2: Comparison with existing works with proposed model

Study	Method	Dataset	XAI	Accuracy
Saravanaprasad (2024)	3D CNN + SVM	Kaggle CT	No	94%
Abe (2025)	Ensemble CNN	IQ-OTH/NCCD	No	98.17%
Haziq (2025)	Sequential CNN	Histopathology	No	95.34%
Emad (2026)	CNN + DNN	LIDC-IDRI	No	98.04%
Rai (2026)	ResNet152 + SHAP	IQ-OTH/NCCD	Yes	97.3%
Proposed	**ResNet18 + Grad-CAM**	**IQ-OTH/NCCD**	**Yes**	**99.00%**

Second, efficient or alternative models like Haziq et al.[33] improve computational efficiency but are evaluated on different data modalities and do not include explainability, limiting their applicability to CT-based diagnosis.

Third, explainable models such as Rai et al.[35] incorporate XAI using SHAP, but use deeper architectures (ResNet152) with higher computational cost and lower accuracy (97.3%).

In contrast, our ResNet18-based model reaches the highest accuracy at 99.00%. It keeps a simple design and includes Grad-CAM to make its predictions easy to see and understand. This gives a good mix of strong performance, low complexity, and clear explainability, which makes it more useful for real-world clinical settings.

4.10 Discussion

The results of the experiments demonstrate the effectiveness of the proposed transfer learning model based on ResNet18 [54] architecture. It effectively performs multi-class lung cancer detection. The model's accuracy score is 99%, which suggests that it can use discriminative features to classify images as normal, benign or cancerous. The model can achieve the efficiency of medical imaging learning by using the pre-trained feature representations, under the premise of data volume is small, by using the transfer learning mechanism, as shown in [75].

The patterns observed in training and validation indicate a steady decline in loss during the course of the experiment, achieving stable convergence after 8 training epochs. This shows that there is no overfitting, thereby confirming the generalizability of the model. Additionally, the use of the AdamW optimizer together with learning rate scheduling helped increase training stability and reduced overtraining possibilities.

Further understanding of the performance of the model was gained through an analysis of the confusion matrix and classification report. It is important to note that in the testing set, all malignant and normal cases were correctly classified by the system. This is important because early detection of malignant cases is essential in treatment planning [2]. The lowest score in terms of recall was observed for the benign class due to one incorrectly classified case. This may occur because benign

and malignant nodules look similar, making it hard to tell them apart on CT scans. Moreover, benign tumors make up fewer examples in the dataset compared to malignant and normal tumors.

Finally, ROC curve analysis confirms that our proposed solution performs well, as all classes achieved very high AUC scores, with the malignant class reaching 1.000, which speaks of excellent separation capacity of the model.

Analysis of Grad-CAM heatmaps revealed that the model is focused only on important parts of the lung image [76]. For example, in malignant tumors, the model identifies parts of the body, where malignant changes occur and can be seen in the CT scans.

When comparing our solution to recently published papers, we should note that our model proved to be comparable in terms of performance while having a much simpler architecture. Existing models require either complex ensembles or hybrid systems to achieve similar results, whereas our lightweight architecture combined with Grad-CAM explainability provides competitive classification results.

4.11 Conclusion

This chapter evaluates the lung cancer classification model's performance using different parameters. The model shows excellent accuracy and sufficient capability of correctly classifying all three classes. The integration of Grad-CAM enhances interpretability, addressing a key limitation of deep learning models. The findings show that the proposed method is accurate and reliable, making it a strong candidate for automated lung cancer detection.

5. Conclusion

5.1 Concluding Remarks

The research aims to develop a deep learning–based model that will classify the CT scan based on whether or not there is cancer in the lungs. Transfer learning will be performed by utilizing a pre-trained ResNet18[54] neural network to classify the lung CT scan into either benign, malignant, or normal.

From the results obtained, the deep learning model performed effectively and was able to achieve an average accuracy rate of 99%. Furthermore, the deep learning model was efficient in identifying cancerous conditions among patients. This is important as it provides valuable information needed for the treatment process. Finally, Grad-CAM was implemented to help users comprehend the sections of the CT scans that contributed to the results obtained.

In summary, our proposed way gives an effective and easy-to-interpret solution for automatic lung cancer detection from CT images.

5.2 Research Contributions

- We developed a deep learning-based system for lung cancer detection. It uses ResNet18 combined with transfer learning to perform multi-class classification.
- This approach demonstrated exceptional efficacy in detection, attaining an accuracy of 99
- The concept of Grad-CAM was implemented to provide visual explanations, overcoming the issue of lack of interpretability encountered in most models.
- Comparison with modern researches was also done, revealing that this model maintains a good trade-off between accuracy, efficiency, and explainability.

5.3 Limitations of the Study

Despite the success that the experiment yields, there are several limitations associated with it:

- Small size of the dataset might hinder generalization by the model.
- There is no evaluation on another dataset outside the one provided for the assessment of the model's performance.
- There are some minor errors made during classification of benign lesions due to imbalanced nature of the classes involved.
- The analysis is purely focused on the use of images without any input from the medical record.

5.4 Future Work

- Using a bigger and more varied dataset to improve the robustness of the algorithm.
- Utilizing MRI images obtained from the clinical environment and developing an algorithm to compare its efficiency with that of CT images.
- Using multimodal data that would include both clinical and demographic data.
- Developing a decision support algorithm that can be used in the clinical setting.

5.5 Summary

To conclude, the current thesis shows that the integration of a deep learning model using transfer learning techniques, along with explainable AI, can achieve accuracy along with increased interpretability for the diagnosis of lung cancer. The suggested technique may help doctors make quick and accurate diagnoses.

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An Explainable Deep Learning with Transfer Learning Approach for Lung Cancer Detection

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